MCM / ICM

A visual language for model evidence

24 figure types + 4 palette comparisons

All values, profiles, locations and scenarios are synthetic demonstrations.

01–04 MODEL & DATA
Workflow • feedback • exploration • comparison

05–09 OPTIMIZATION
Pareto • 3D surface • parallel axes • feasible domain • response

10–14 ALLOCATION & SENSITIVITY
Sankey • budget mix • correlation • tornado • Morris

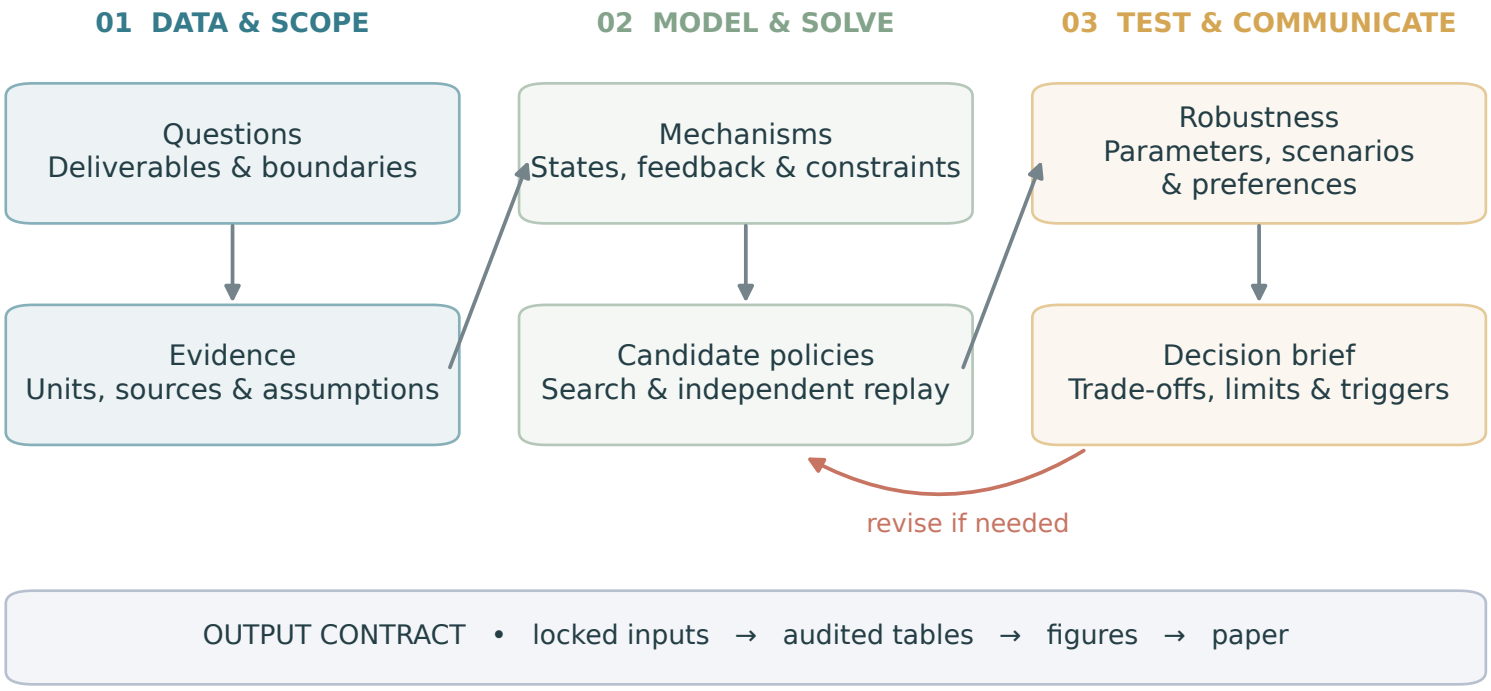
15–18 UNCERTAINTY & DECISIONS
Distributions • fan chart • regret • policy regimes

19–24 SPACE & VALIDATION
Network • transfers • radar • residuals • margins • convergence

25–28 COLOR STUDIES
Coastal • Academic • Terracotta • High contrast

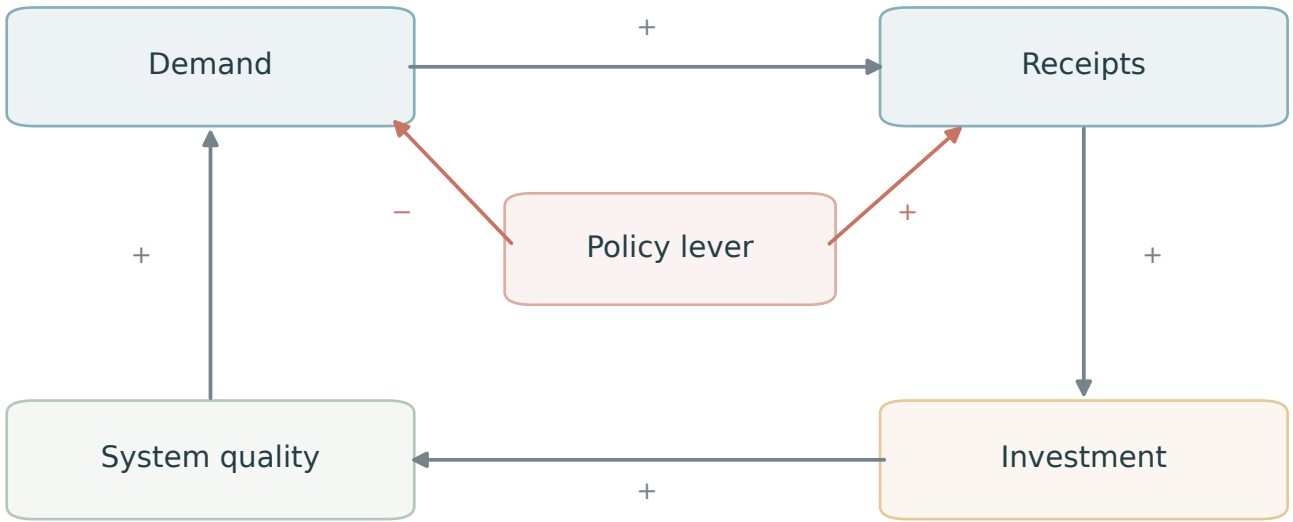
From questions to defensible recommendations

A modular workflow with explicit checks and decision outputs



A policy feedback system

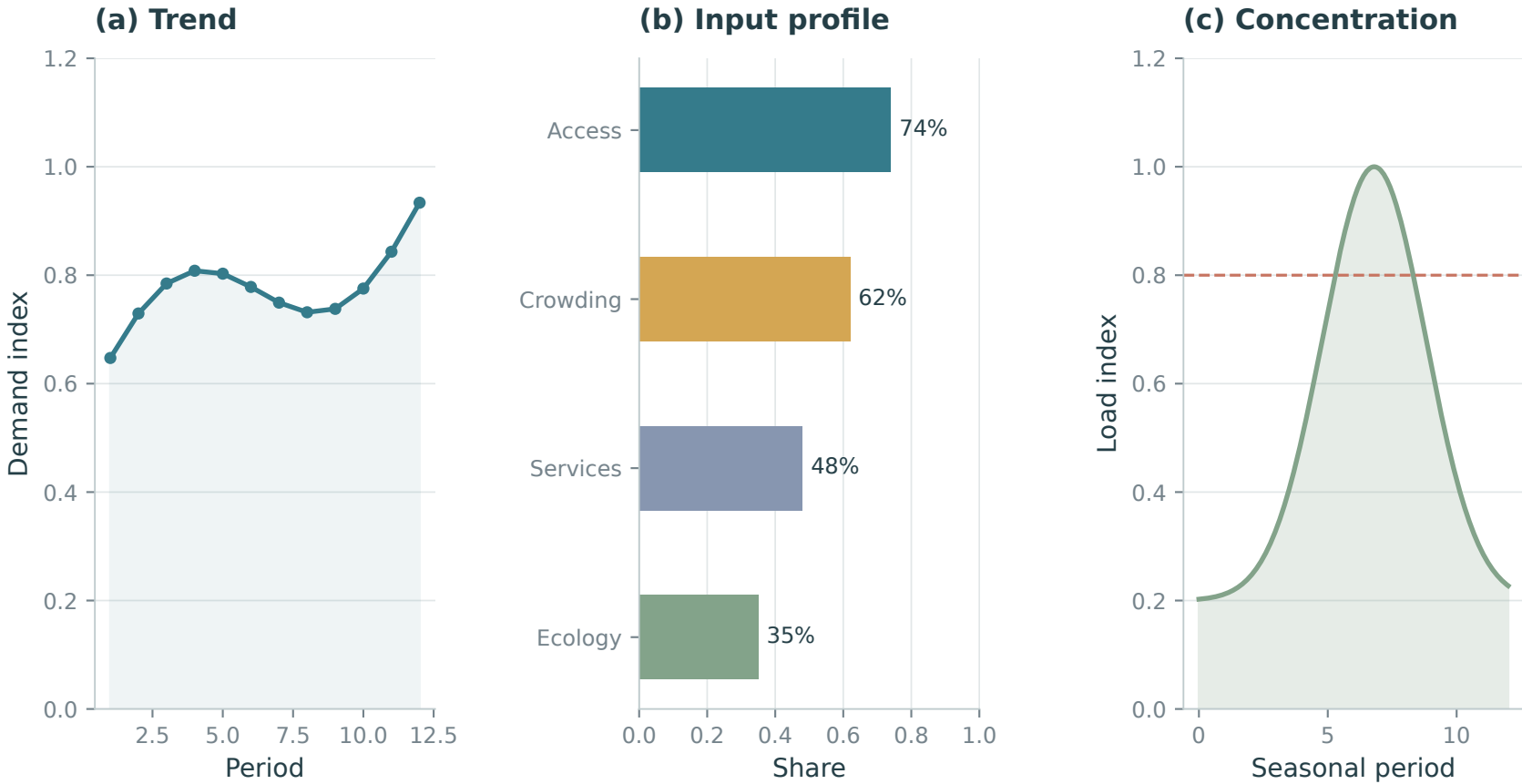
Signed arrows show local directions of influence in a conceptual loop



Investment feedback • directional policy effects

Read the data before solving the model

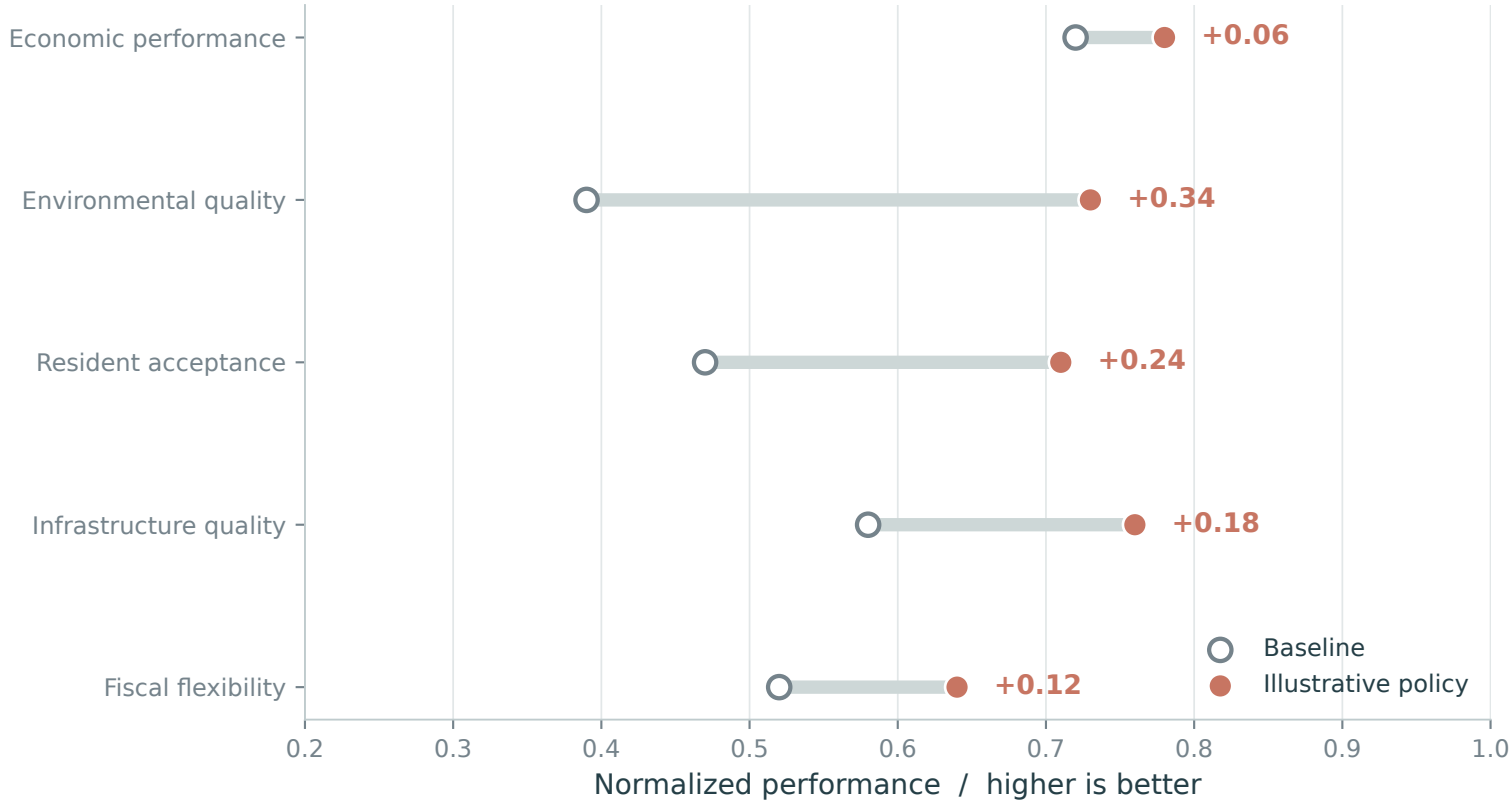
Three complementary views of an illustrative input dataset



SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

A direct comparison of policy outcomes

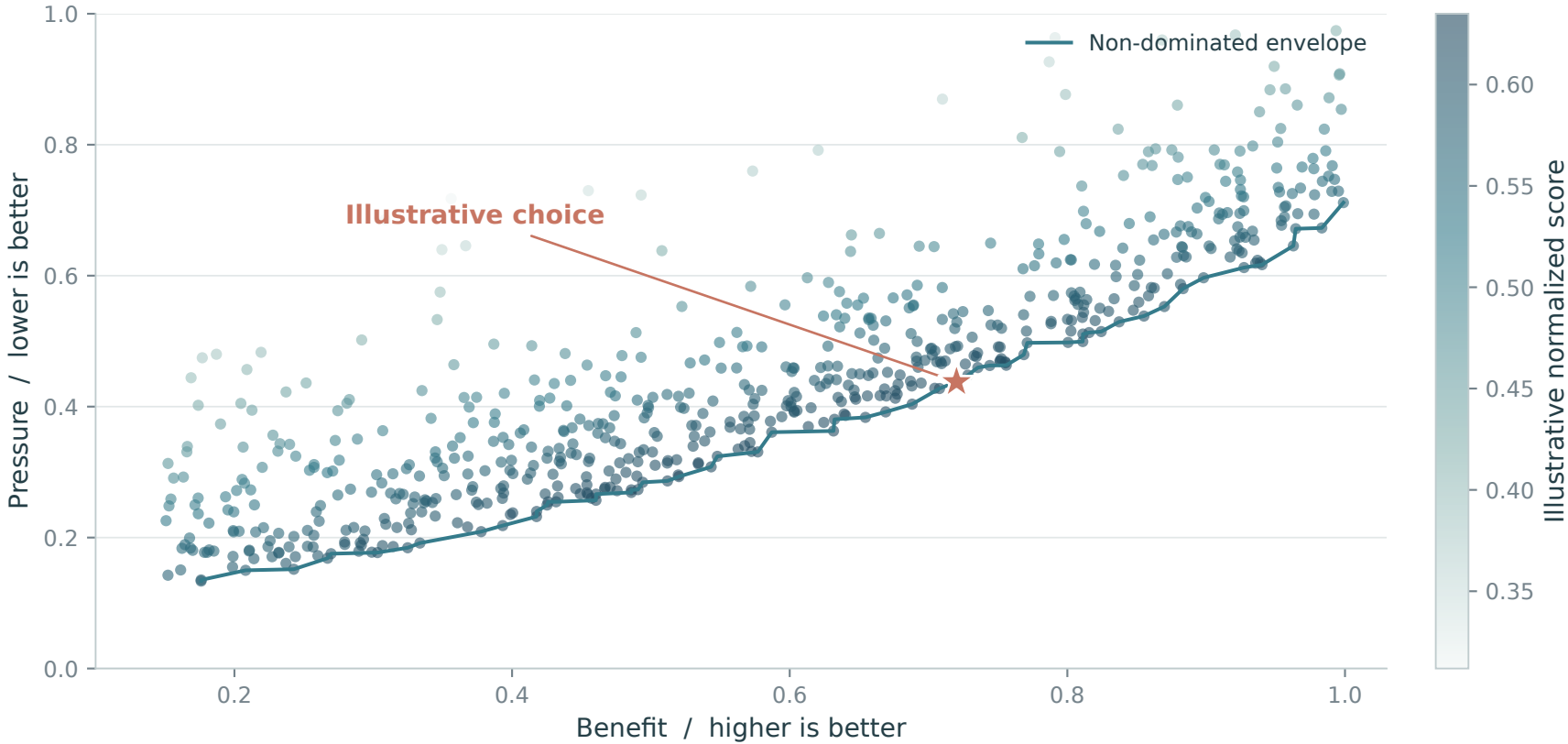
Connected dots preserve scale and reveal the direction of each change



SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

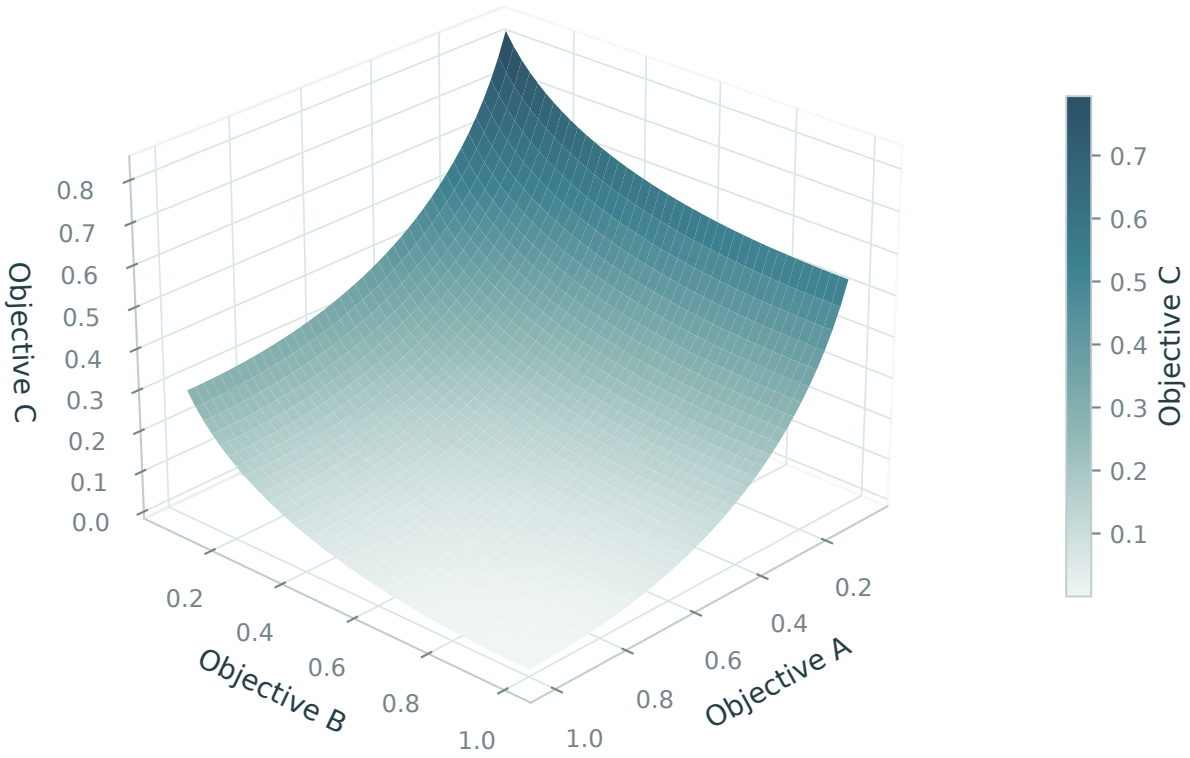
A trade-off that can be read at a glance

The envelope is computed from all generated candidate policies



A three-objective Pareto surface

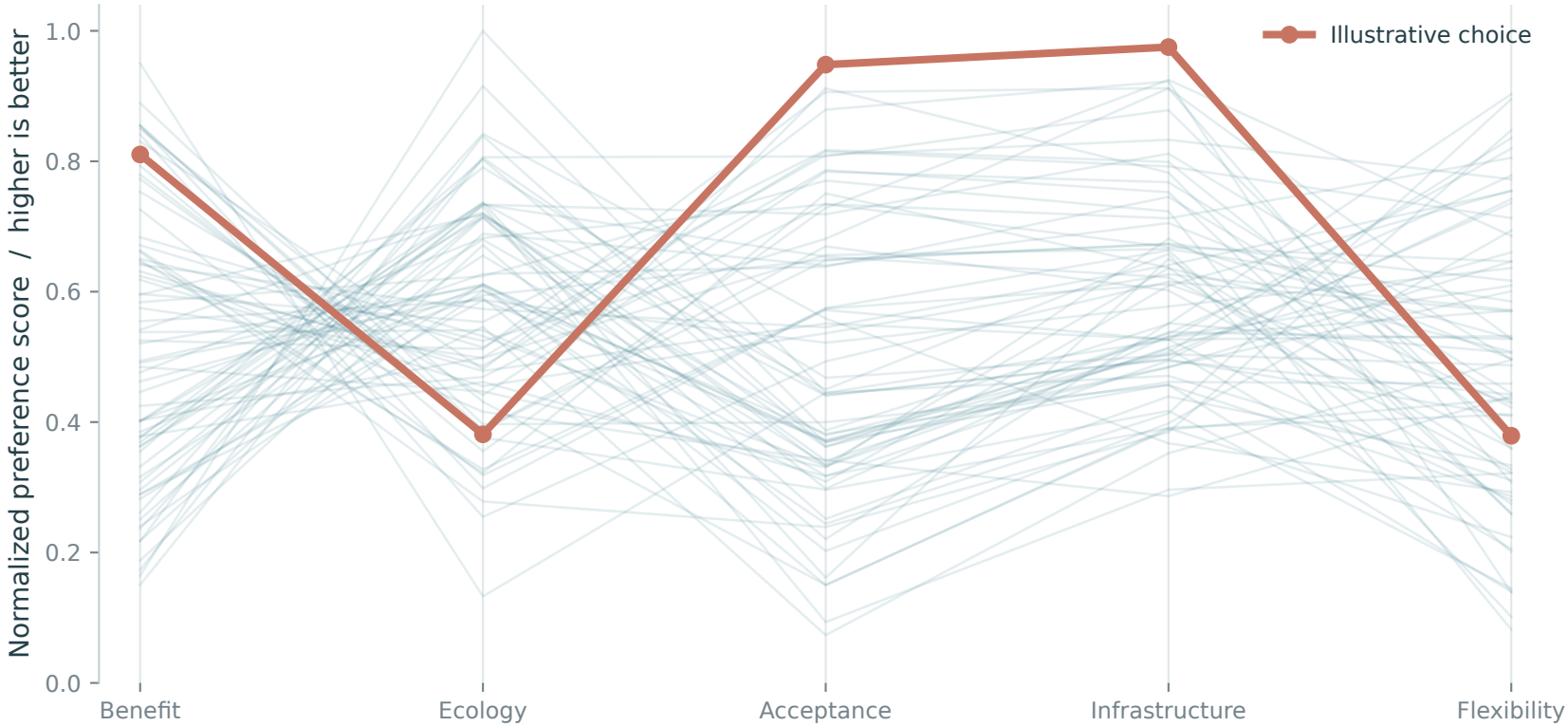
An explicit analytic surface with three distinct objectives; all are minimized



SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

Read several objectives together

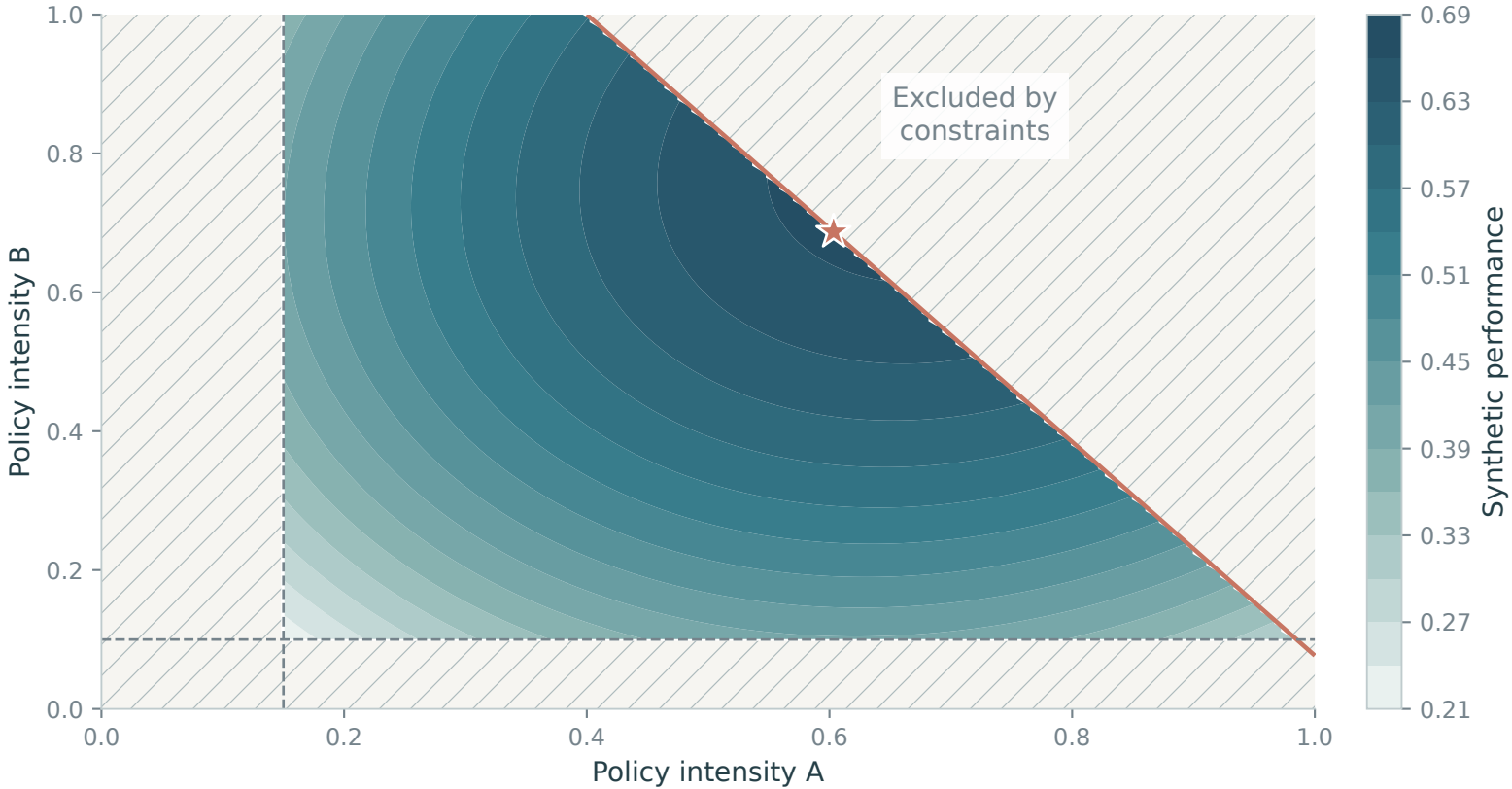
Every axis is normalized to a common preference direction



SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

Show the actual constraint boundary

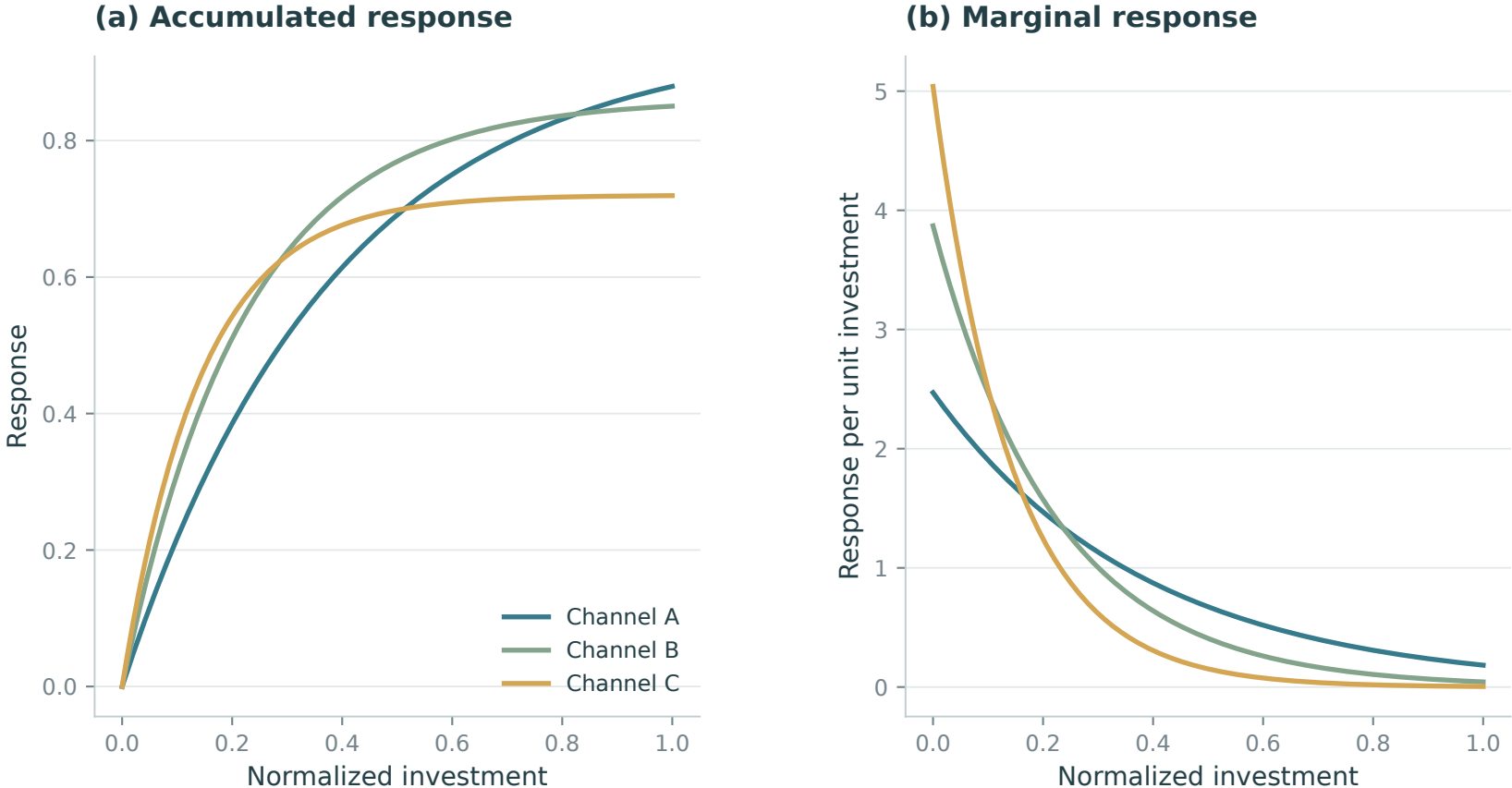
Colors encode performance; hatching marks the excluded part of the grid



SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

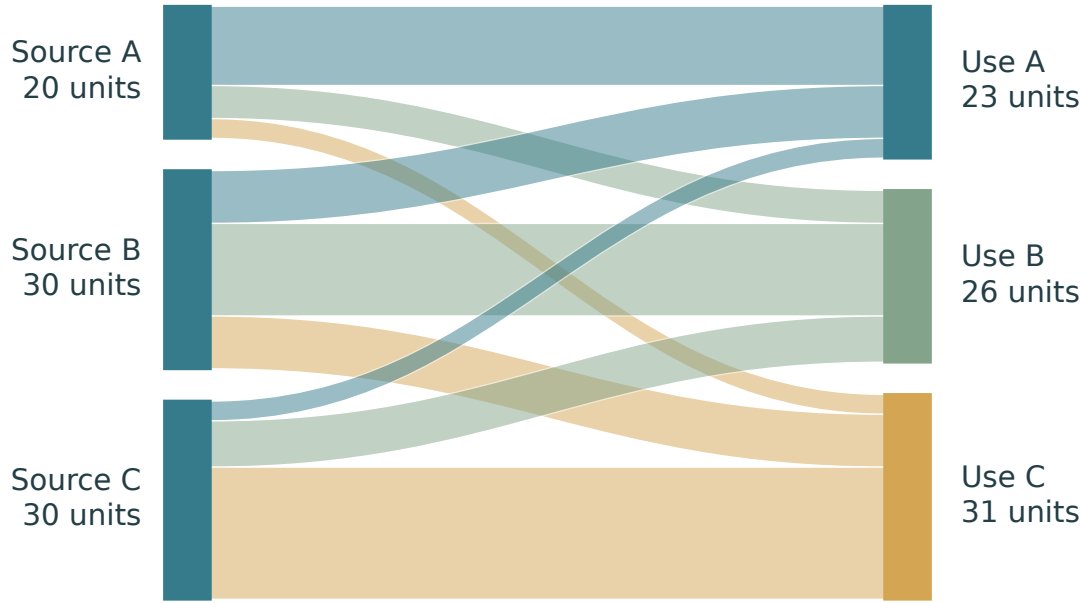
Total response and marginal return

The same analytic response is read in levels and derivatives



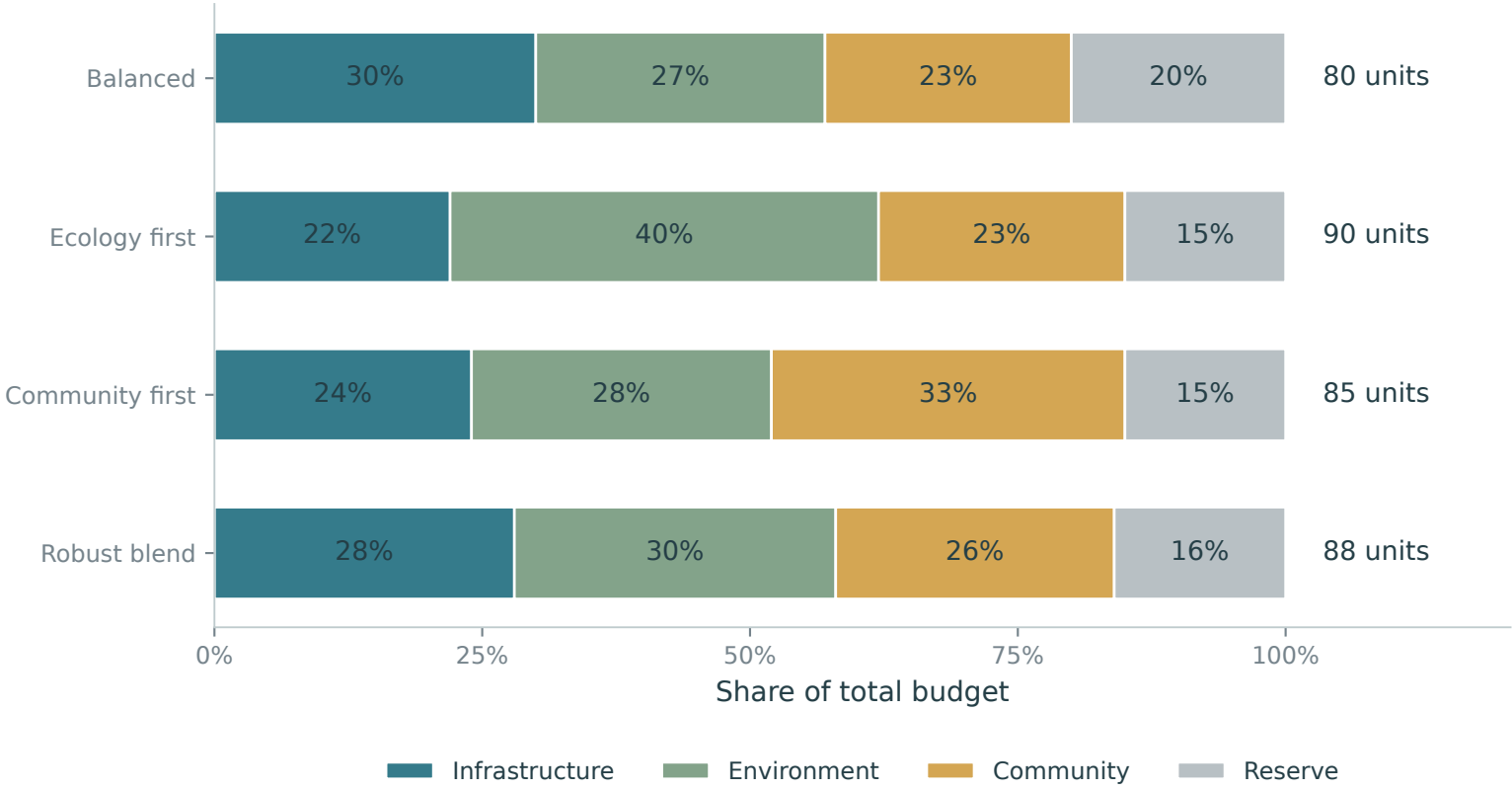
Follow every unit of the budget

Ribbon widths are proportional to a conserved synthetic allocation



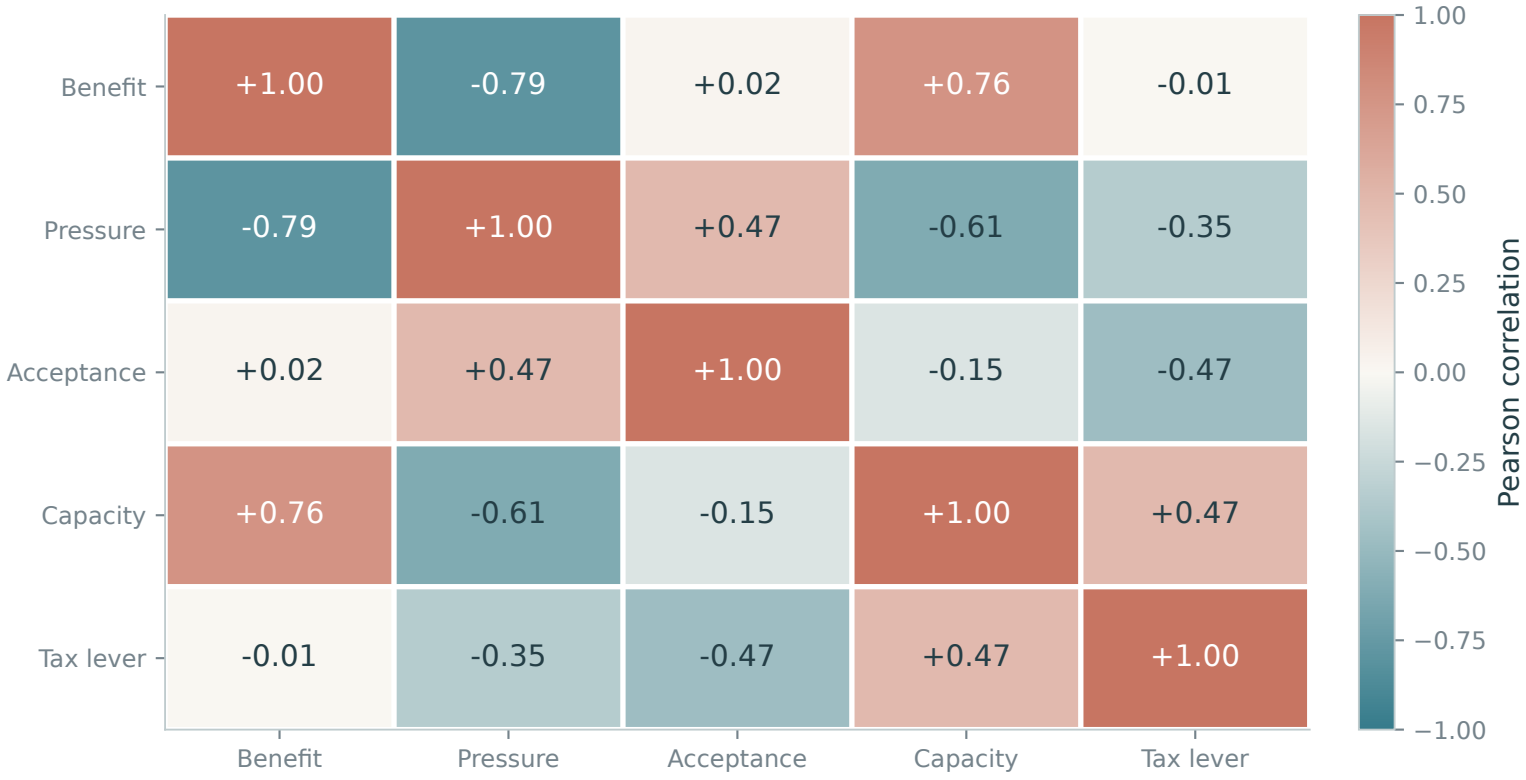
How the allocation changes by strategy

Composition and absolute totals are displayed separately



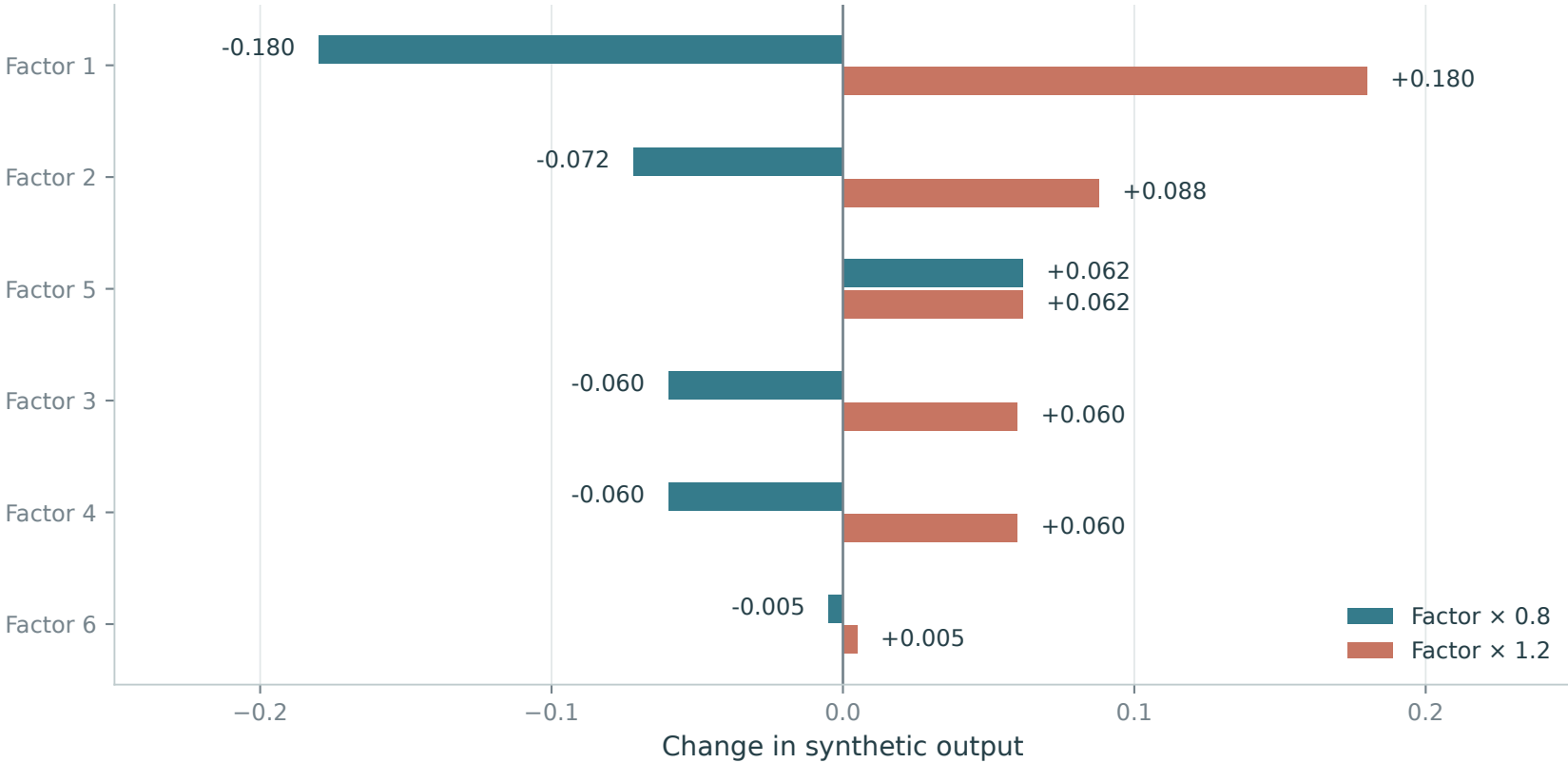
Which model quantities move together?

A signed correlation matrix from a generated multivariate sample



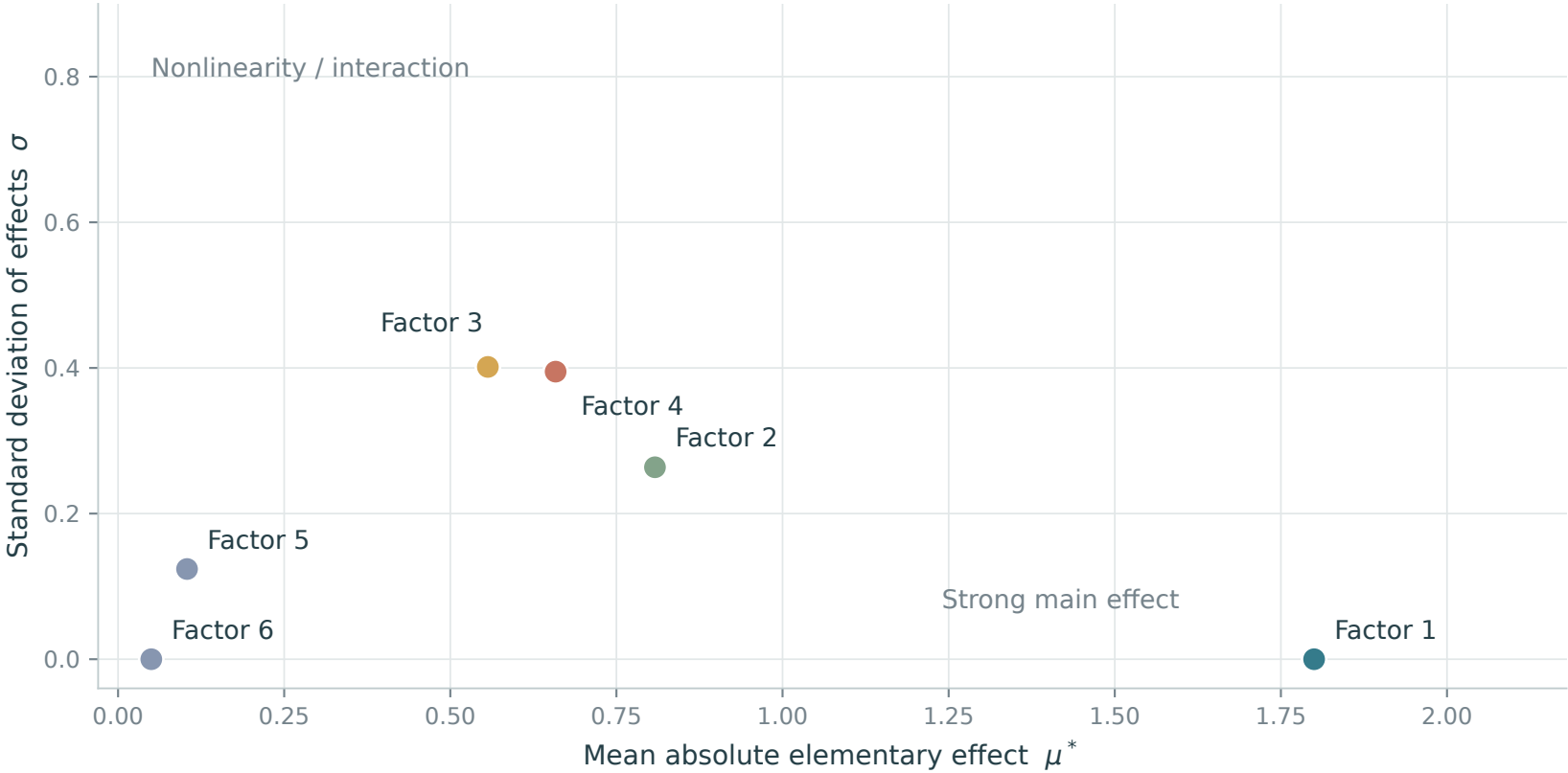
Local sensitivity around a common baseline

Each factor changes by ±20%; all others stay fixed



Separate influence from nonlinear response

Morris elementary effects computed on a six-factor toy function; 160 trajectories

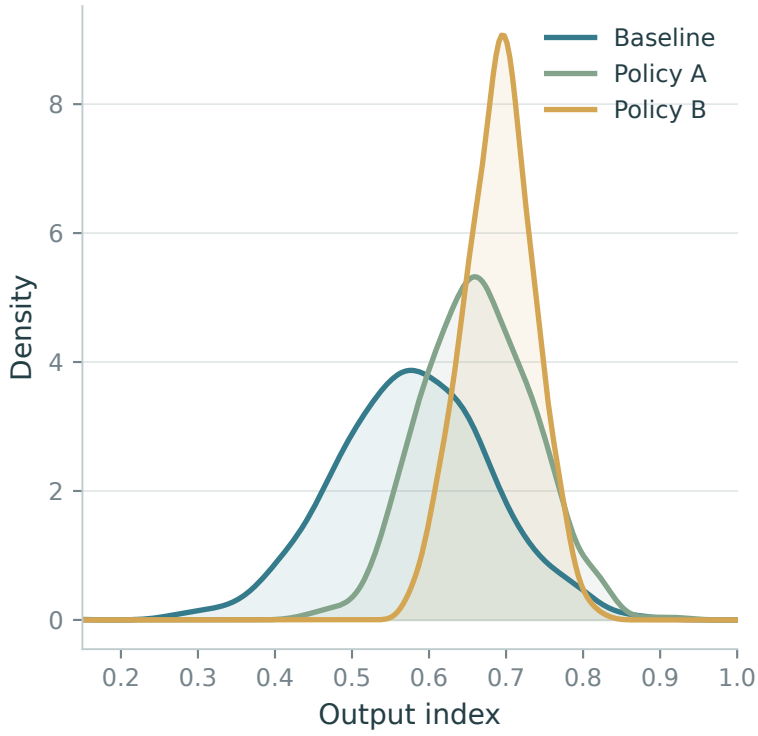


SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

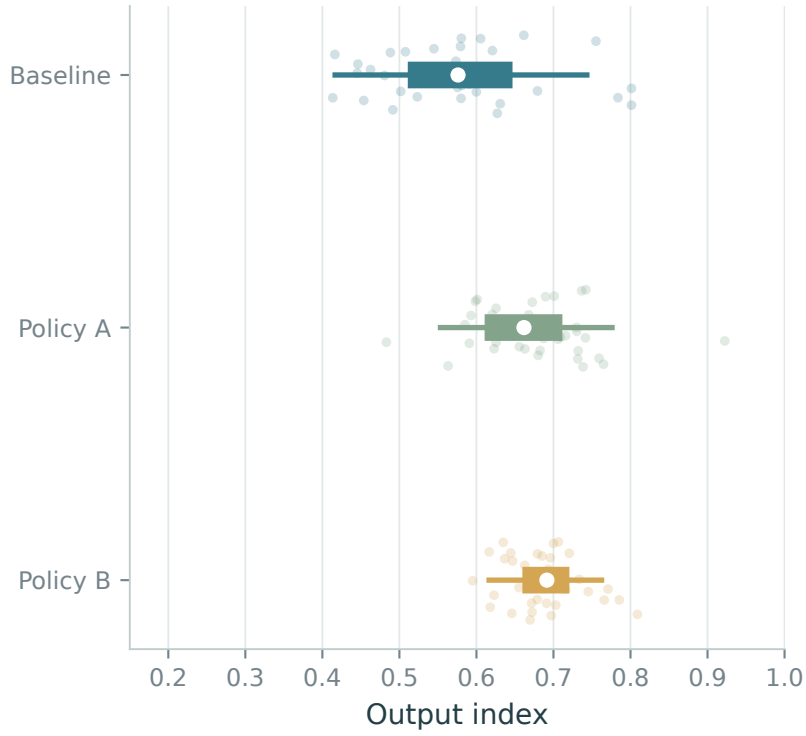
Distribution, spread and tail behavior

Generated samples are summarized by densities, observations and quantile intervals

(a) Estimated density



(b) Median and intervals

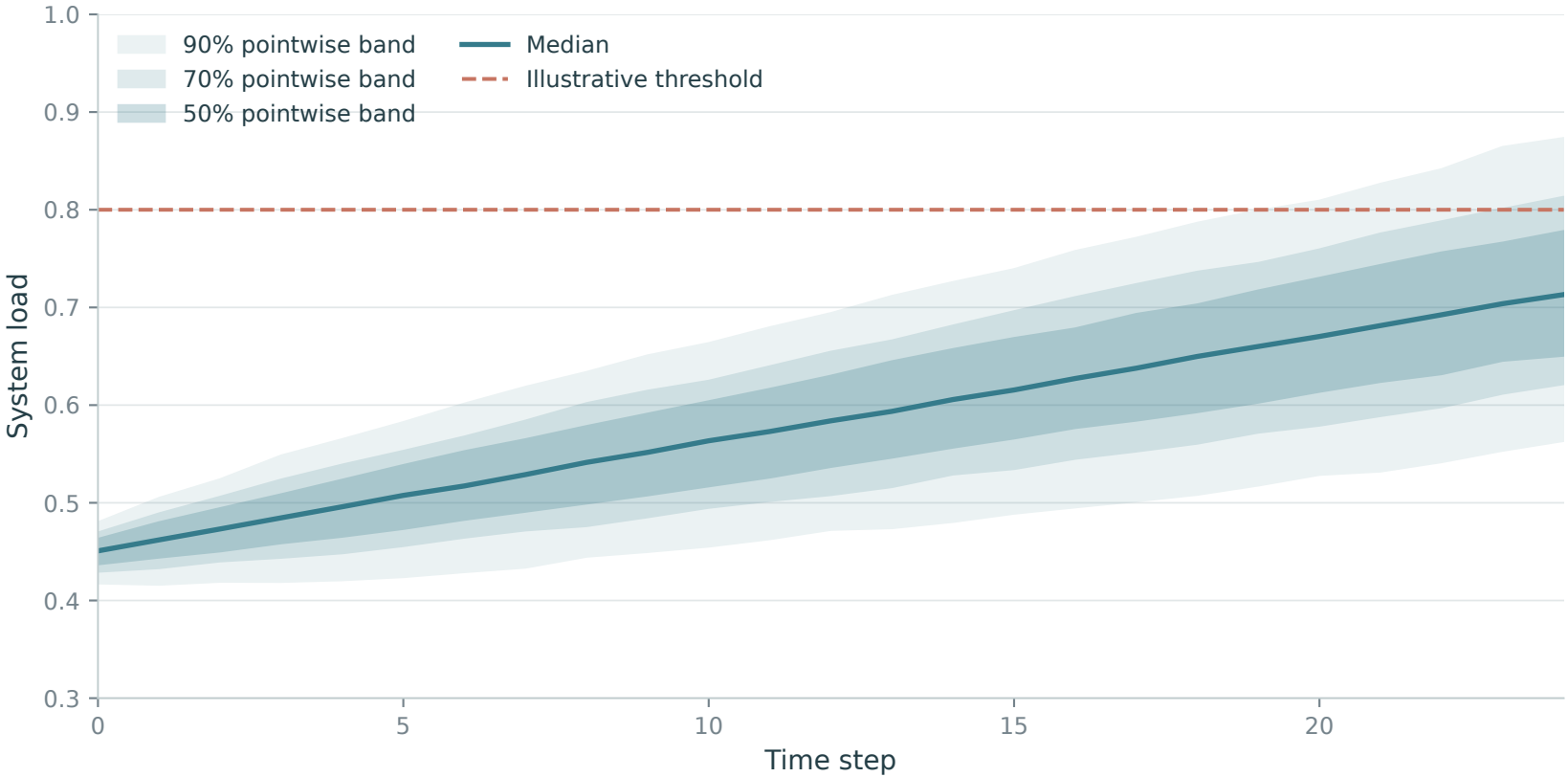


Thick 50% / thin 90% / dot median

SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

Uncertainty through time

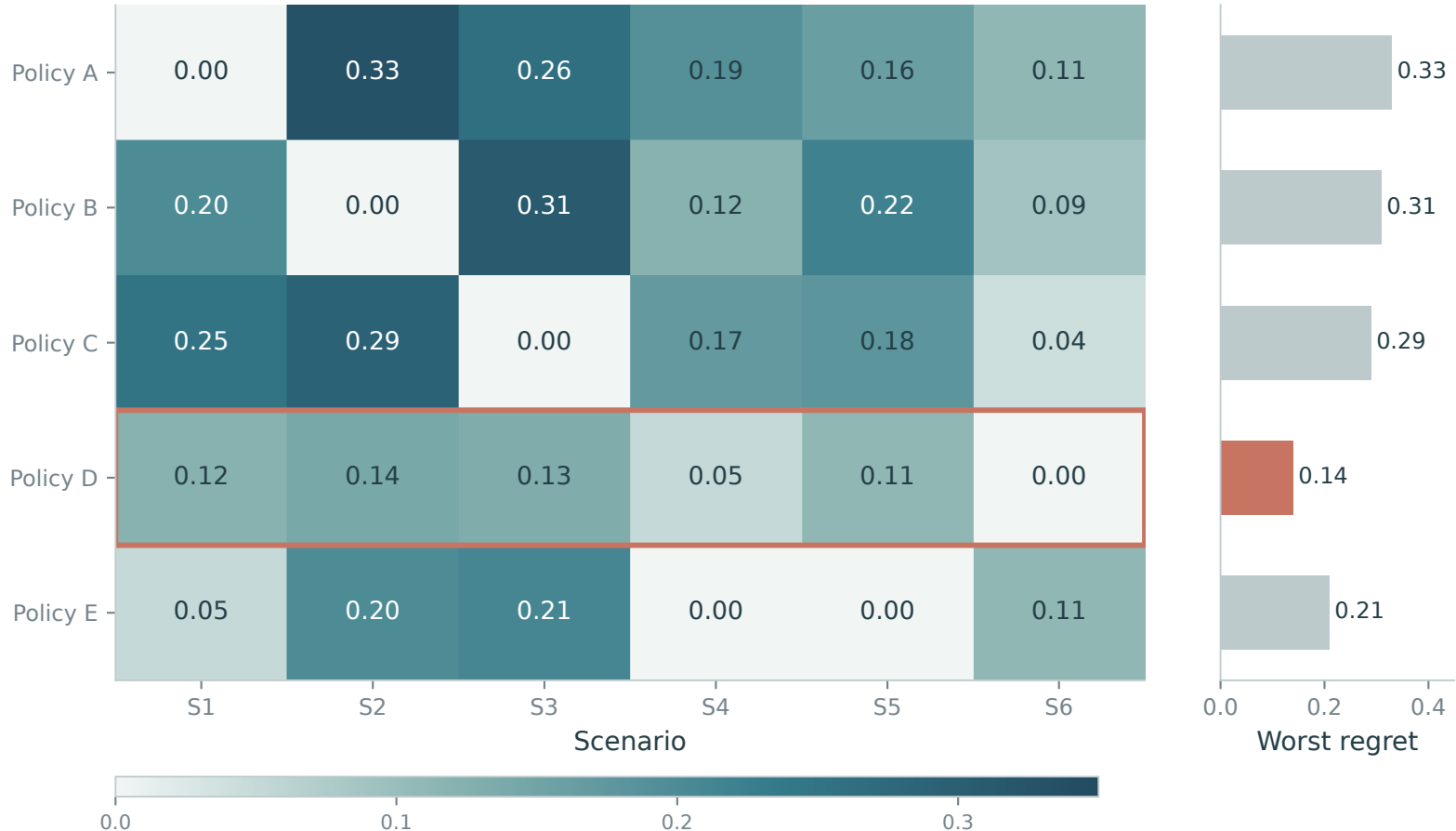
Nested bands are pointwise sample quantiles from 1,500 synthetic paths



SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

A policy that limits the worst regret

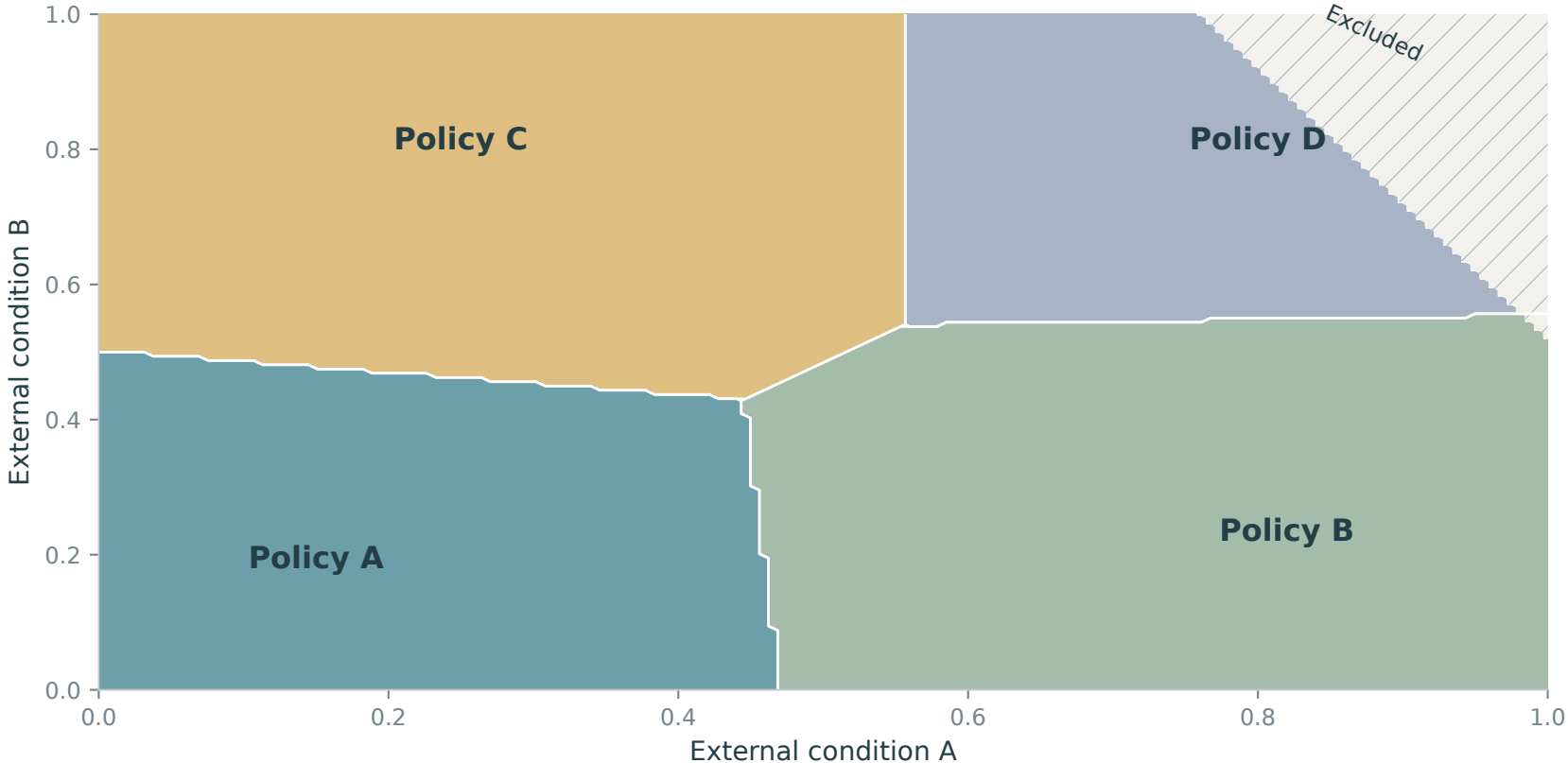
Regret is the gap to the best policy in each scenario; lower is better



SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

Where does the preferred strategy change?

Each region selects the largest of four explicit synthetic policy scores

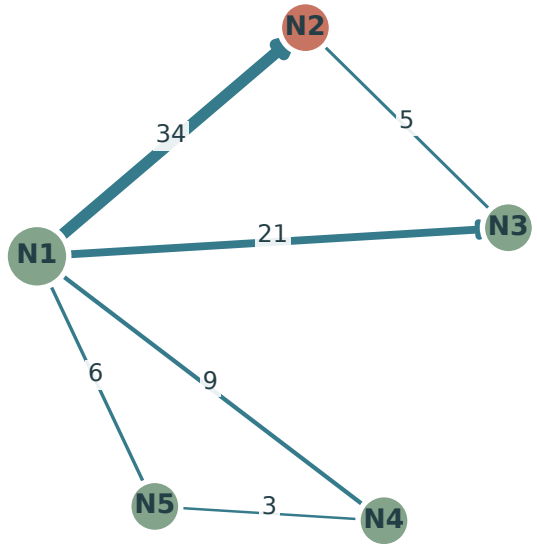


SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

Redistribute flow across a network

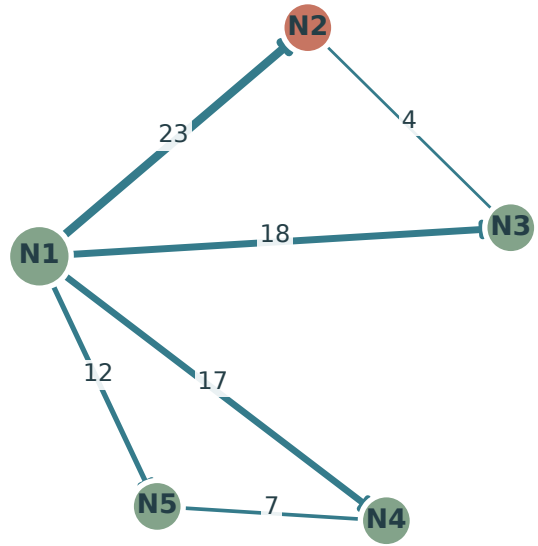
Node positions are schematic; edge widths encode generated flow volumes

(a) Concentrated allocation



SCHEMATIC POSITIONS / NO GEOGRAPHIC COORDINATES

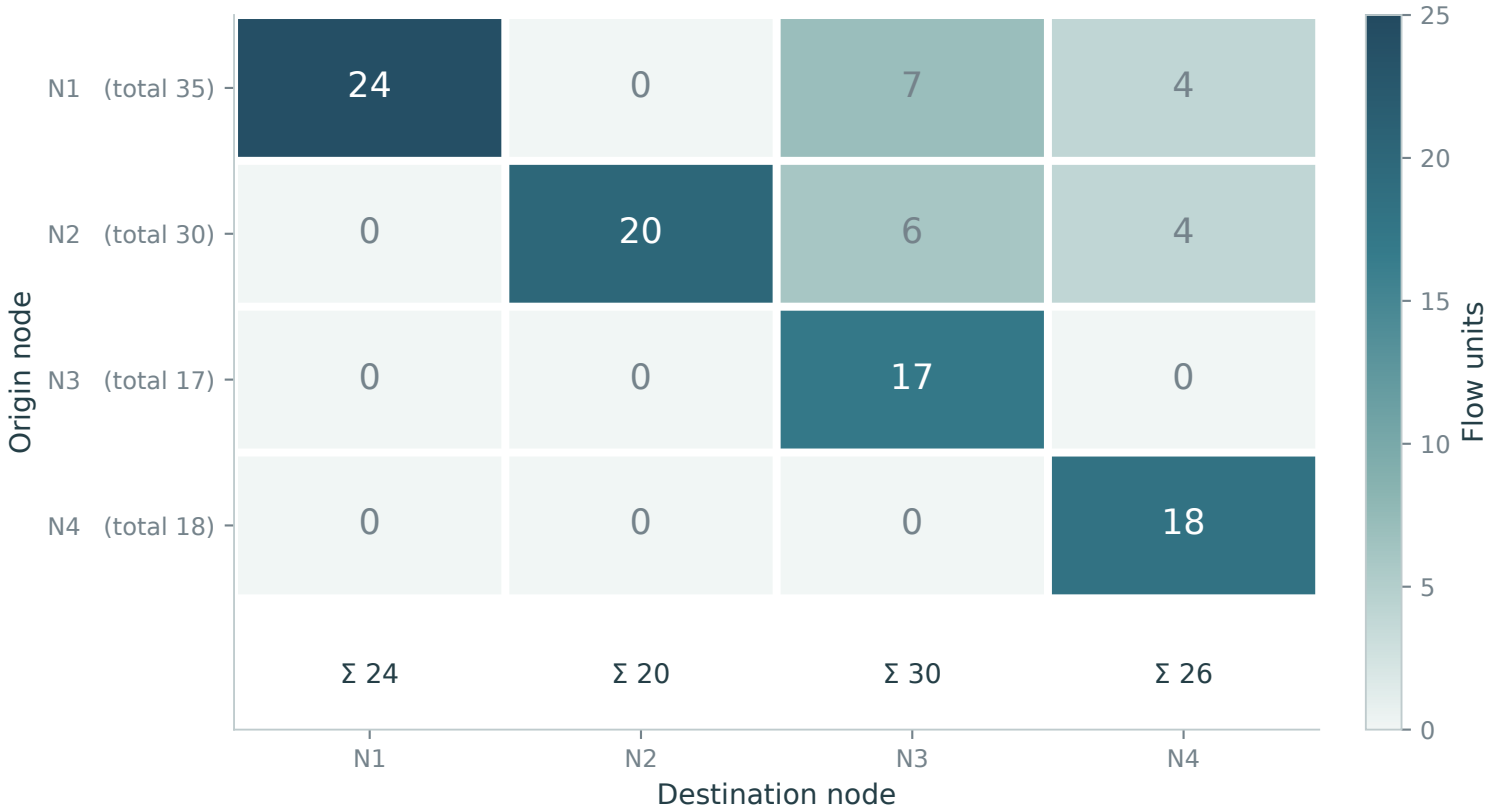
(b) Distributed allocation



SCHEMATIC POSITIONS / NO GEOGRAPHIC COORDINATES

Account for every transfer

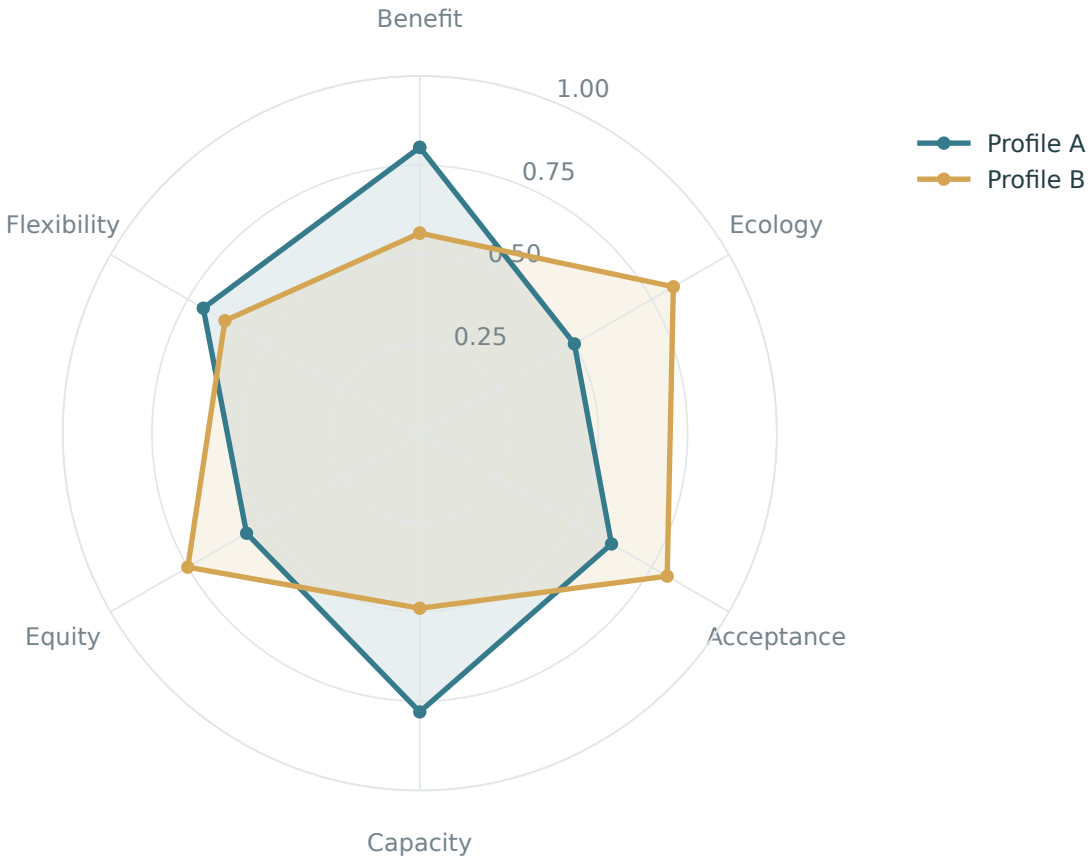
Rows preserve origin totals; columns reveal the new allocation



SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

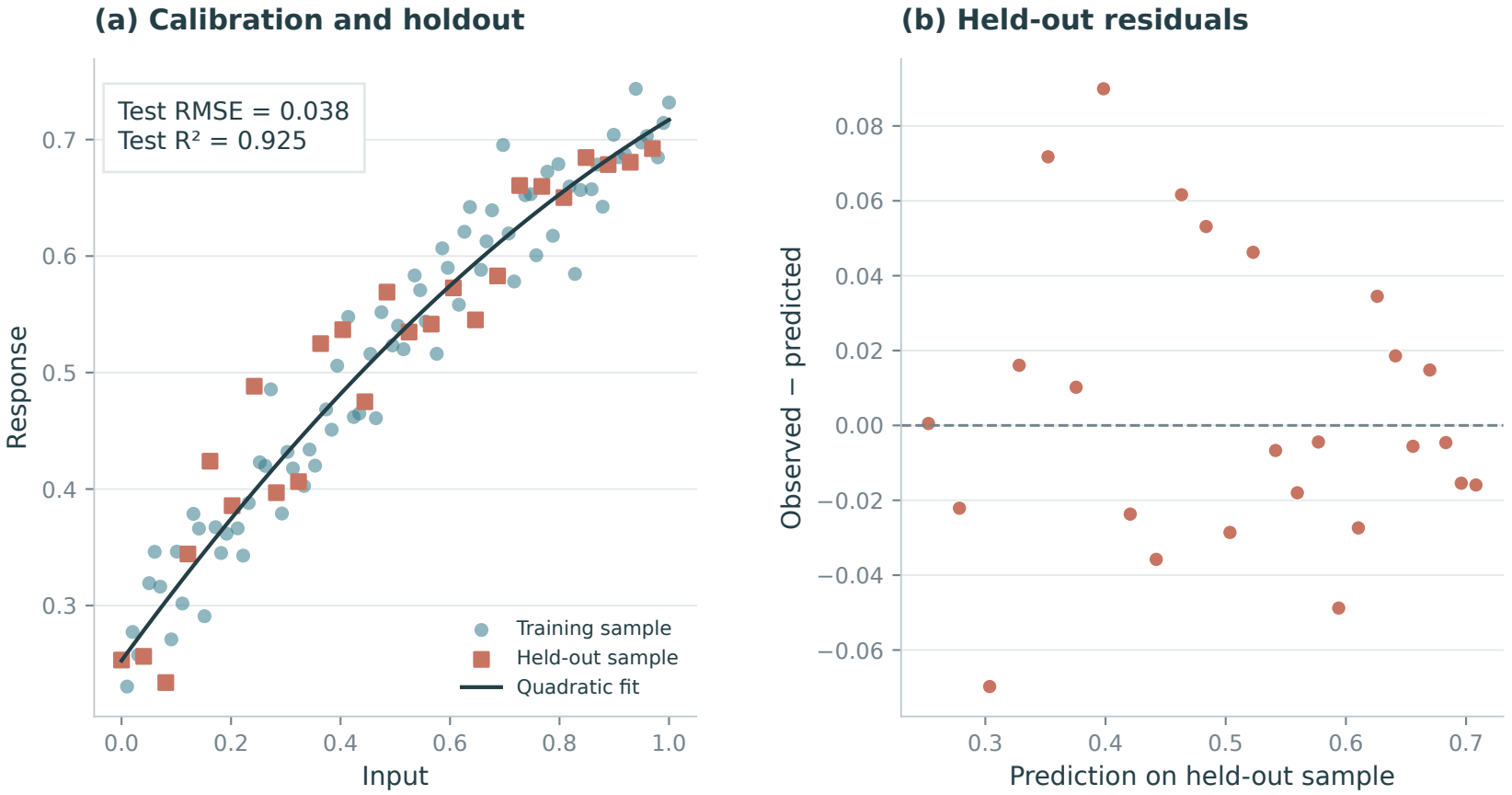
Two contrasting multidimensional profiles

All axes use a common 0–1 scale; profiles are compared axis by axis



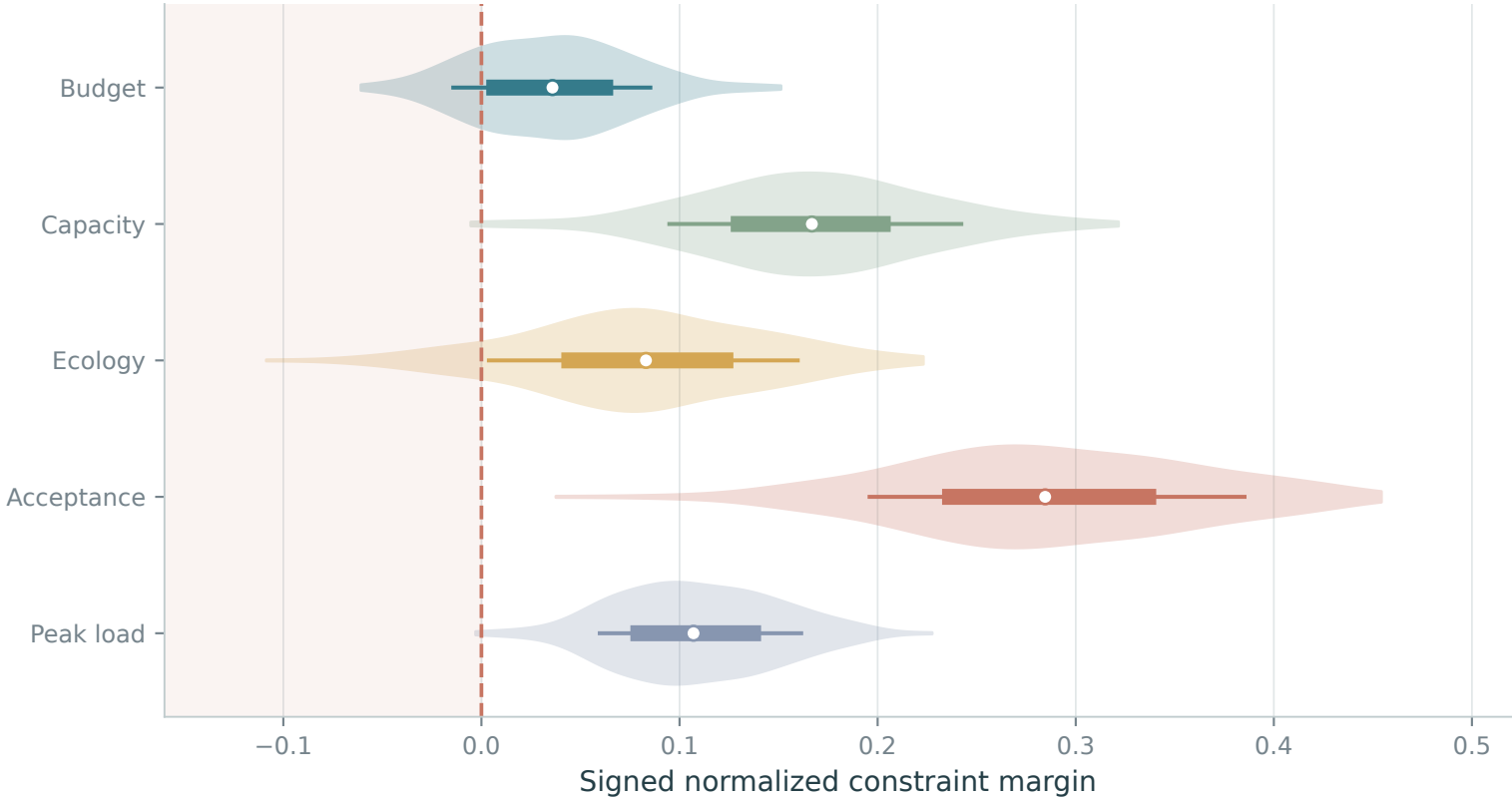
Fit, hold out, and inspect the errors

A generated nonlinear relationship with a held-out test set



Which constraints are close to binding?

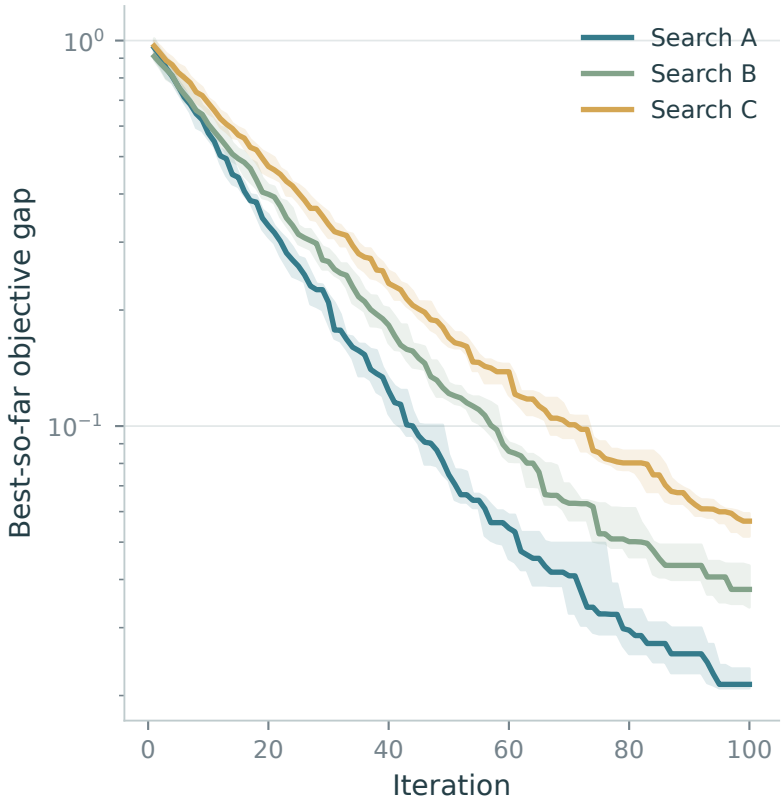
Signed normalized margins from generated policies; negative values indicate violations



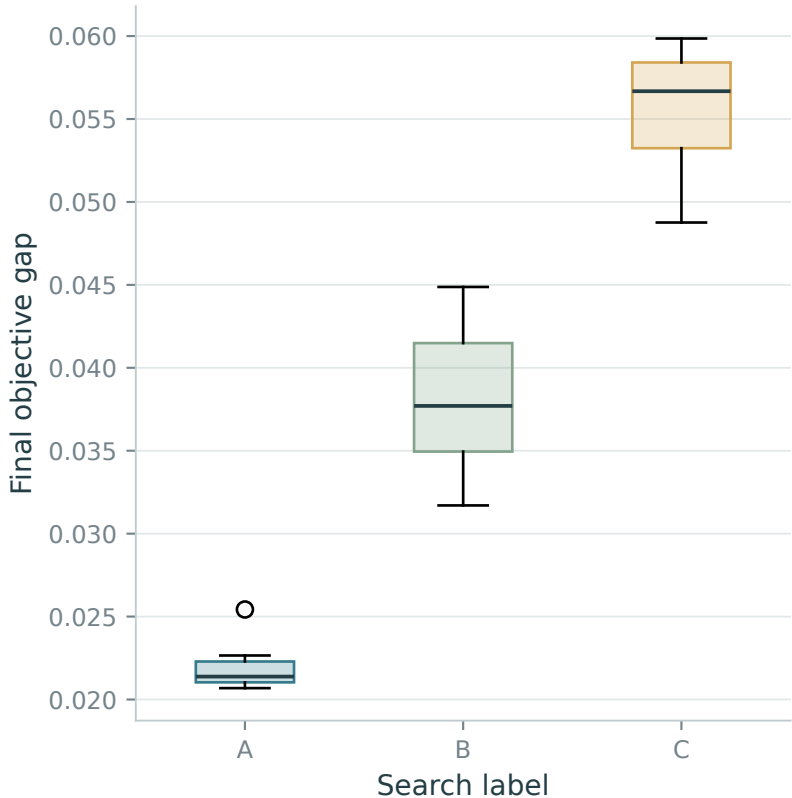
Report search behavior and run-to-run variation

These generated traces illustrate a reporting format for solver diagnostics

(a) Median and 10–90% band



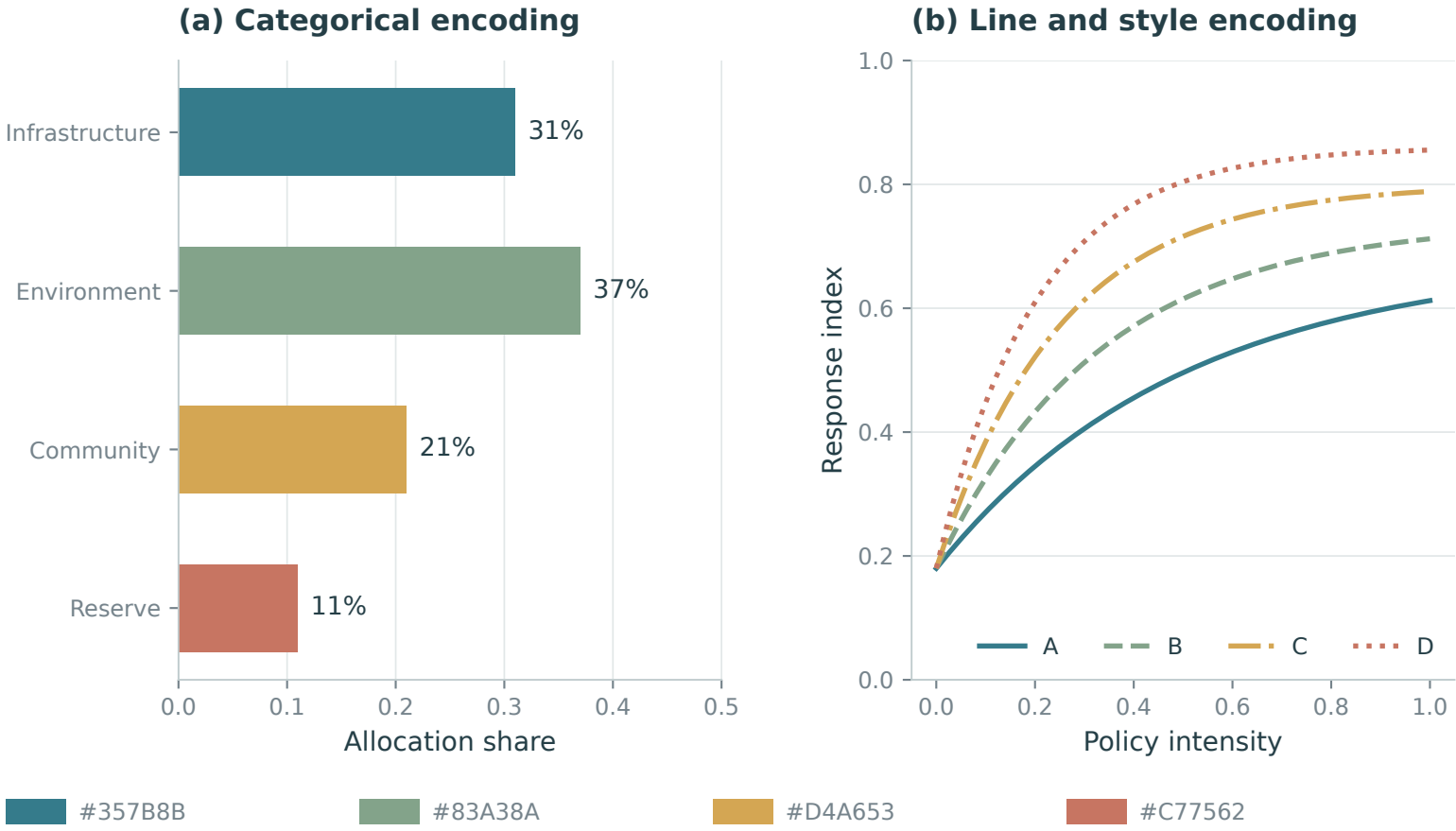
(b) Eight runs per search



SYNTHETIC DEMO / Normalized or arbitrary units / No field observations

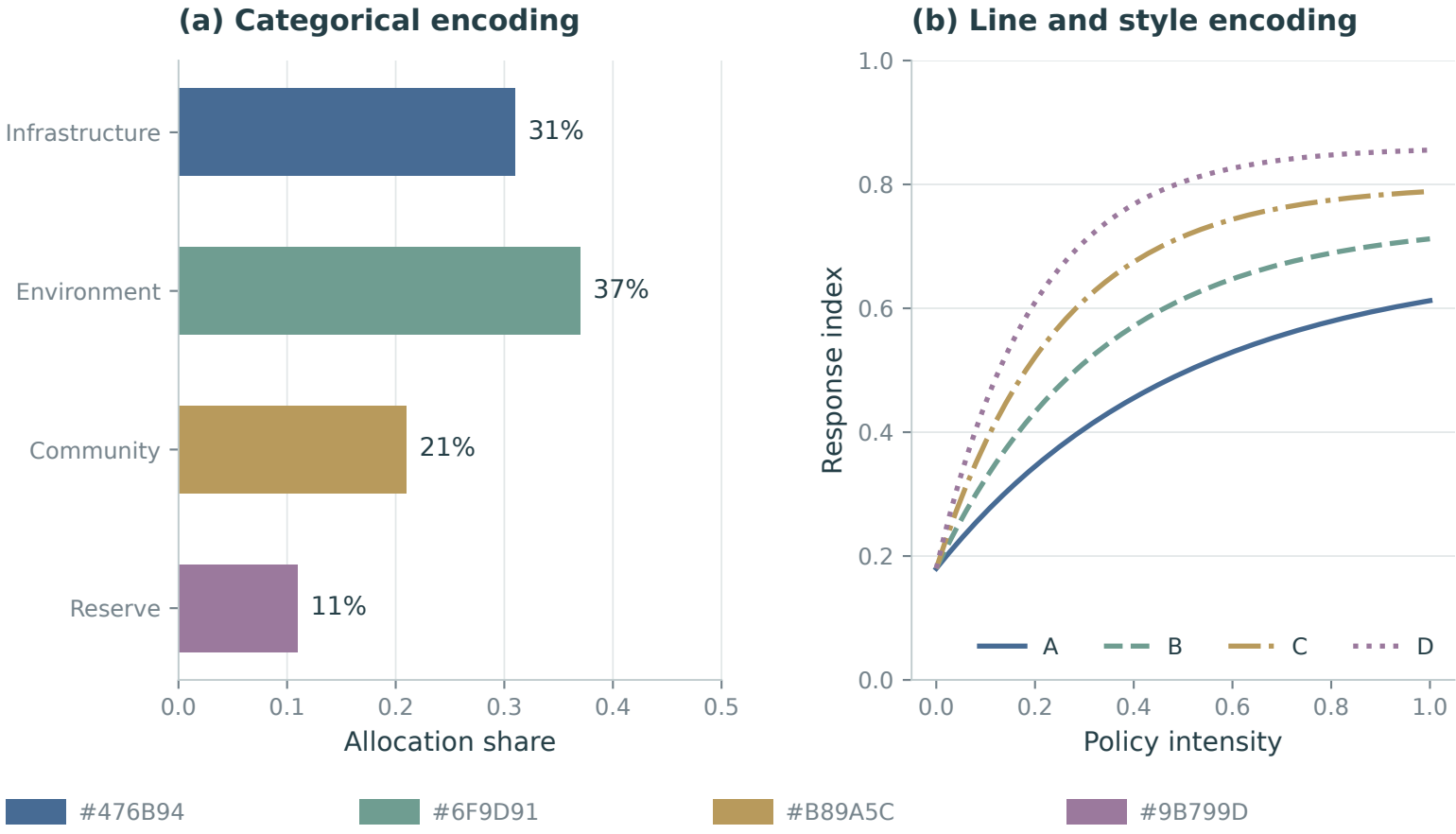
Coastal — the same data, a different palette

Identical geometry, scales and values in all four color studies



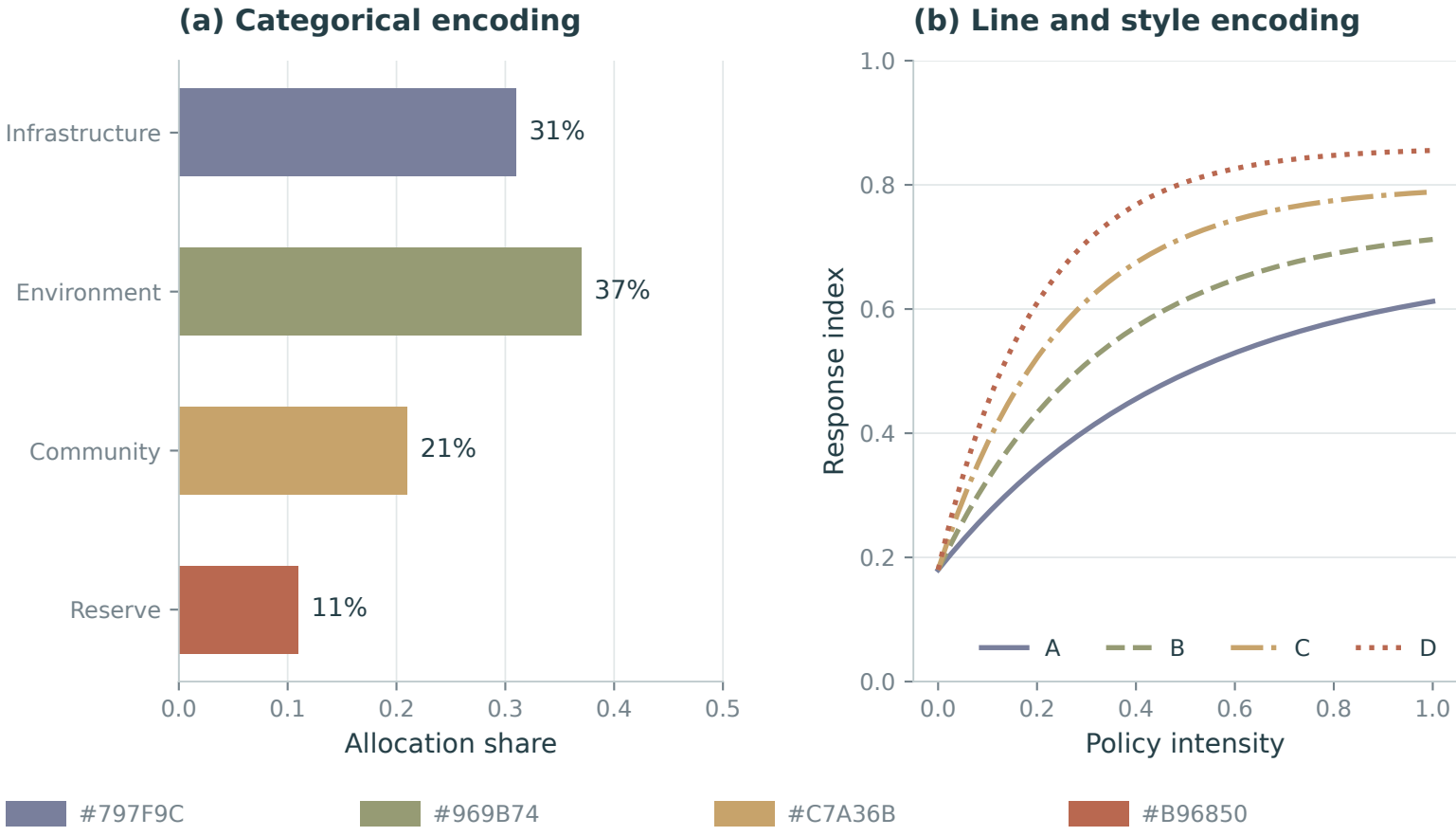
Academic — the same data, a different palette

Identical geometry, scales and values in all four color studies



Terracotta — the same data, a different palette

Identical geometry, scales and values in all four color studies



High contrast — the same data, a different palette

Identical geometry, scales and values in all four color studies

